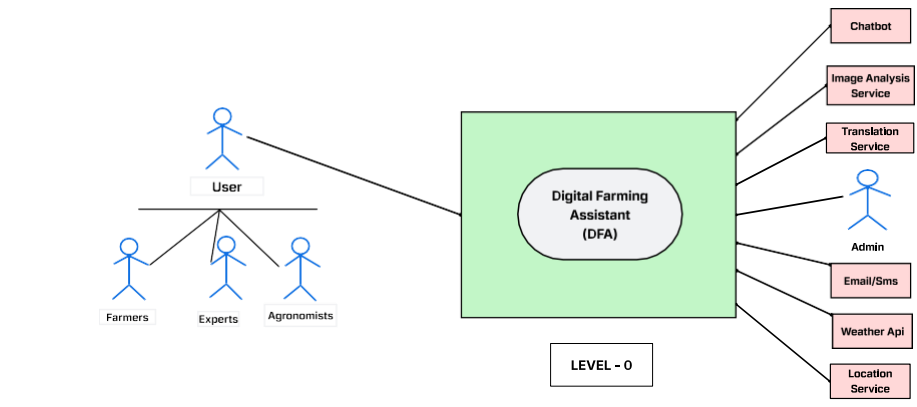
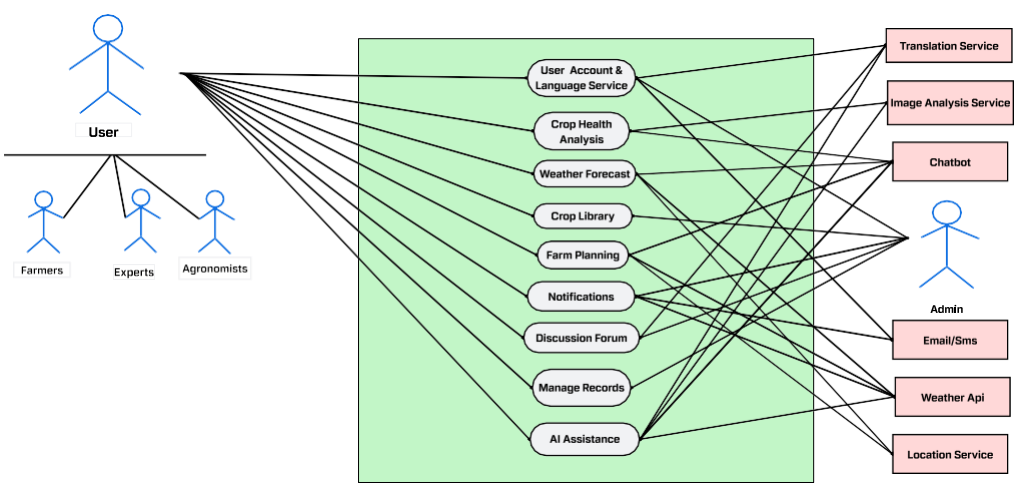
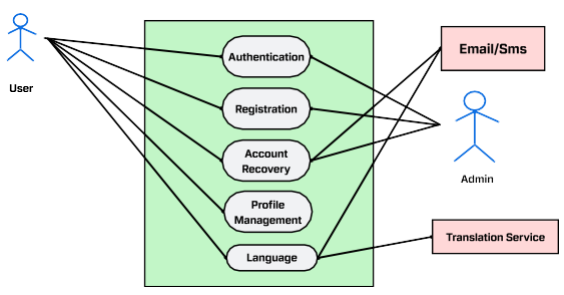


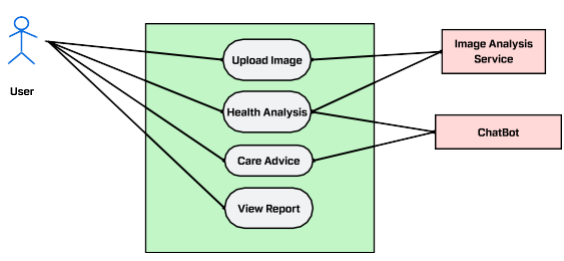
LEVEL - 0



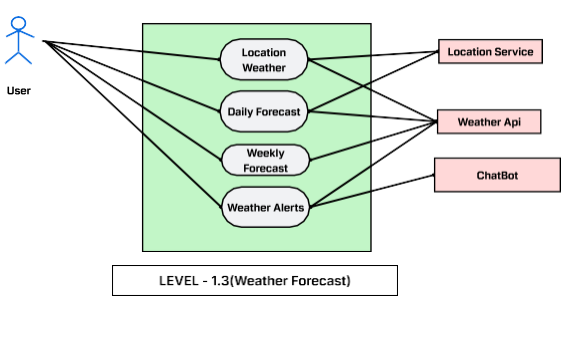
LEVEL - 1 (DFA)



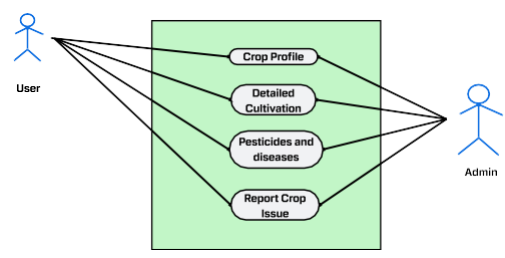
LEVEL - 1.1 (User Account & Language Service)



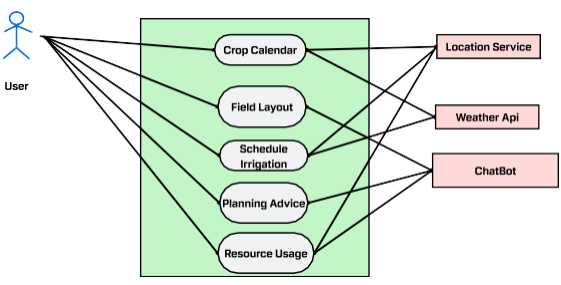
LEVEL - 1.2(Crop Health Analysis)



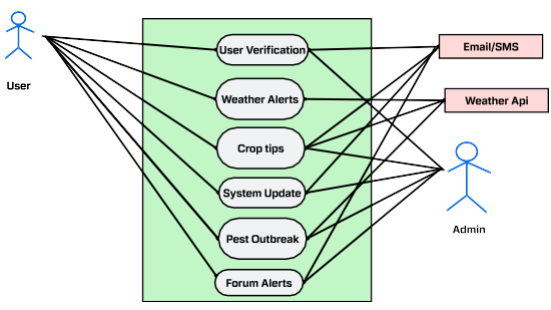
LEVEL - 1.3(Weather Forecast)



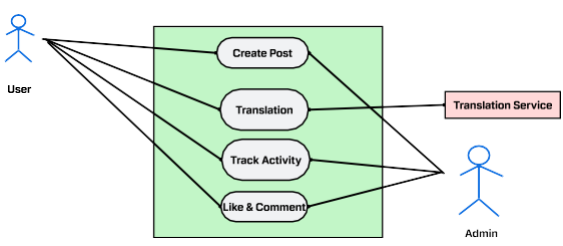
LEVEL - 1.4(Crop Library)



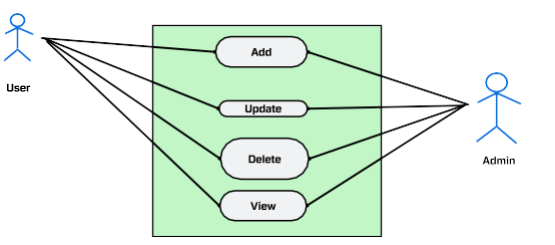
LEVEL - 1.5(Farm Planning)



LEVEL - 1.6(Notifications)



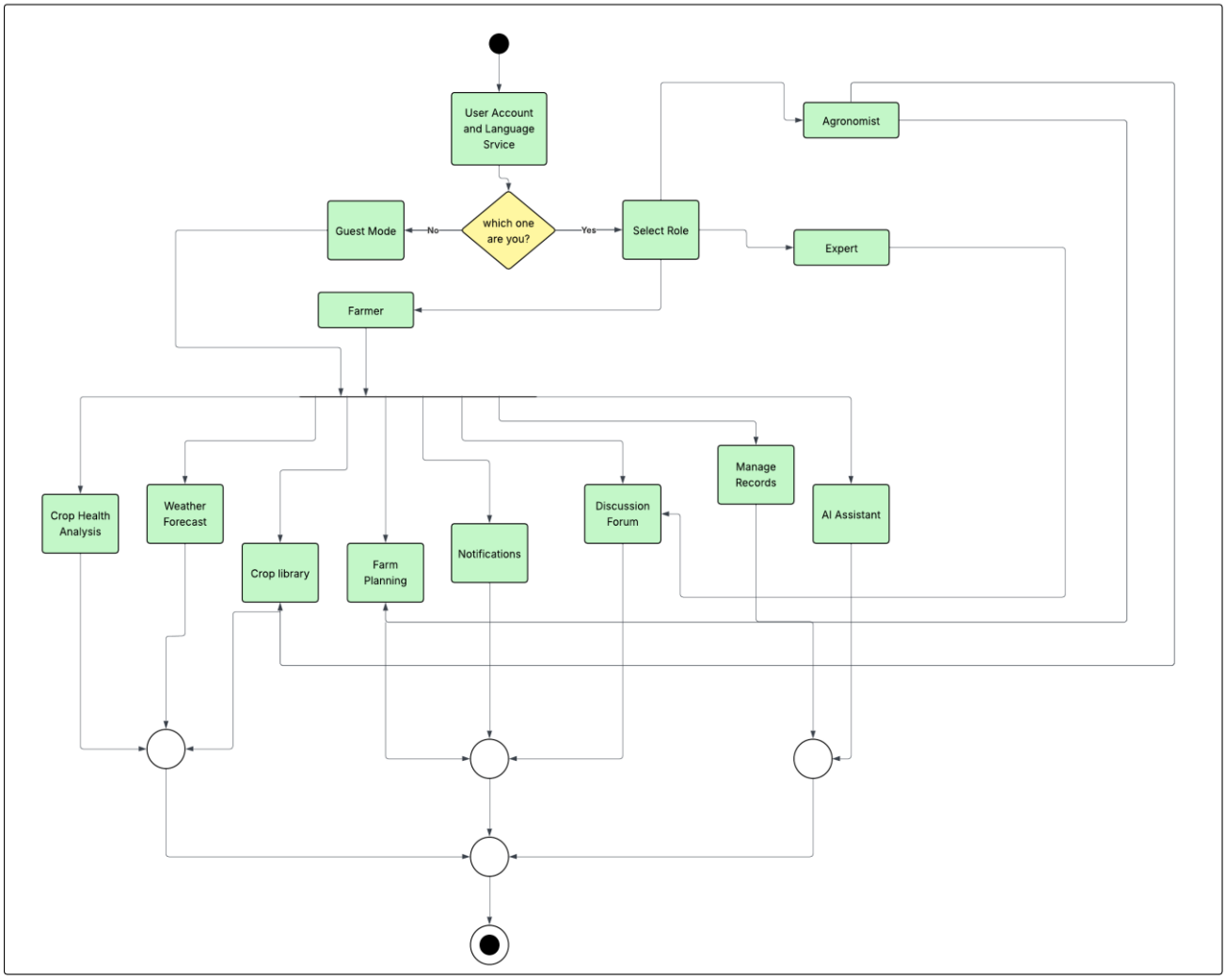
LEVEL - 1.7(Discussion Forum)



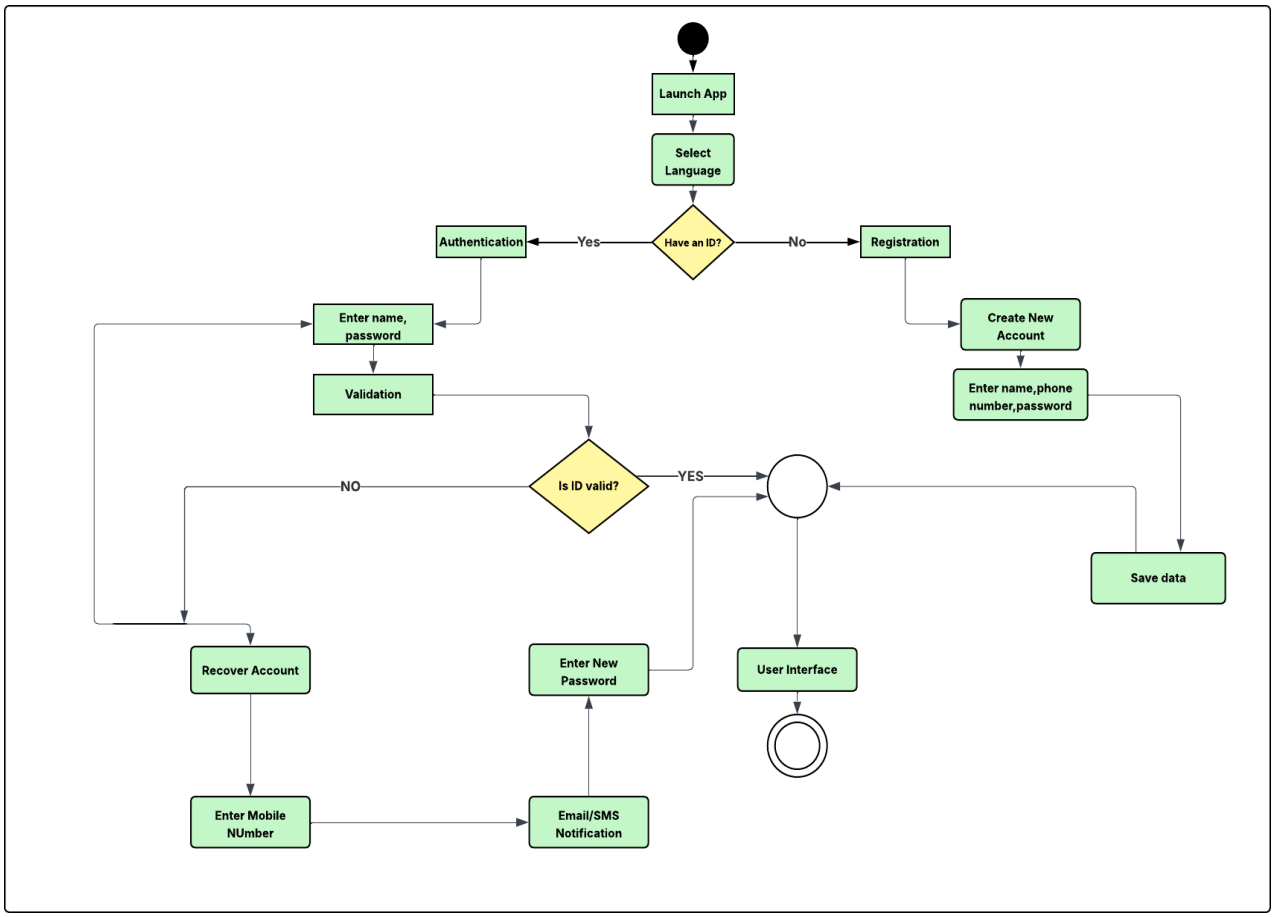
LEVEL - 1.8(Manage Records)

LEVEL - 1.9(AI Assistance)

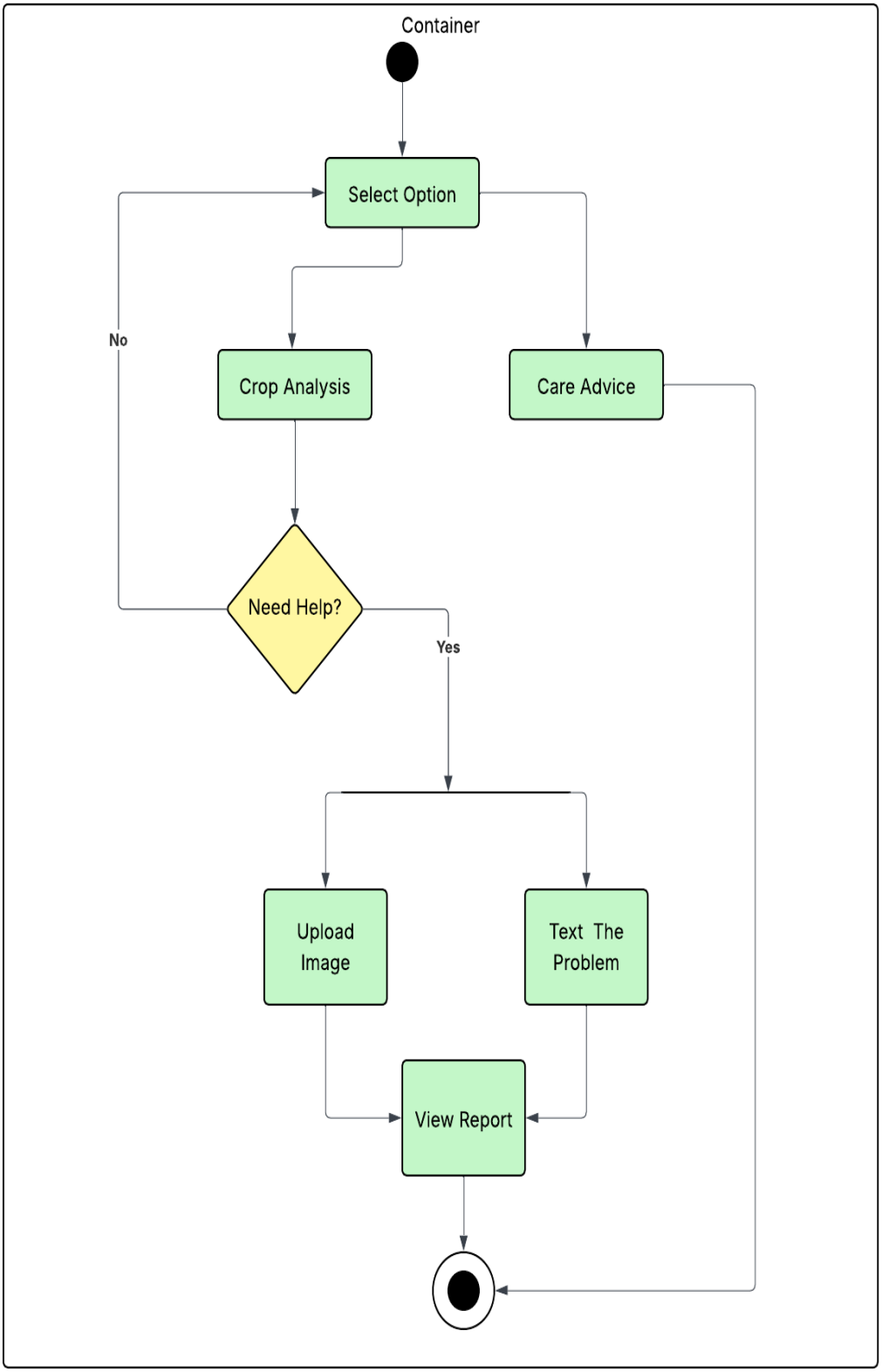
Digital Farming Assistant



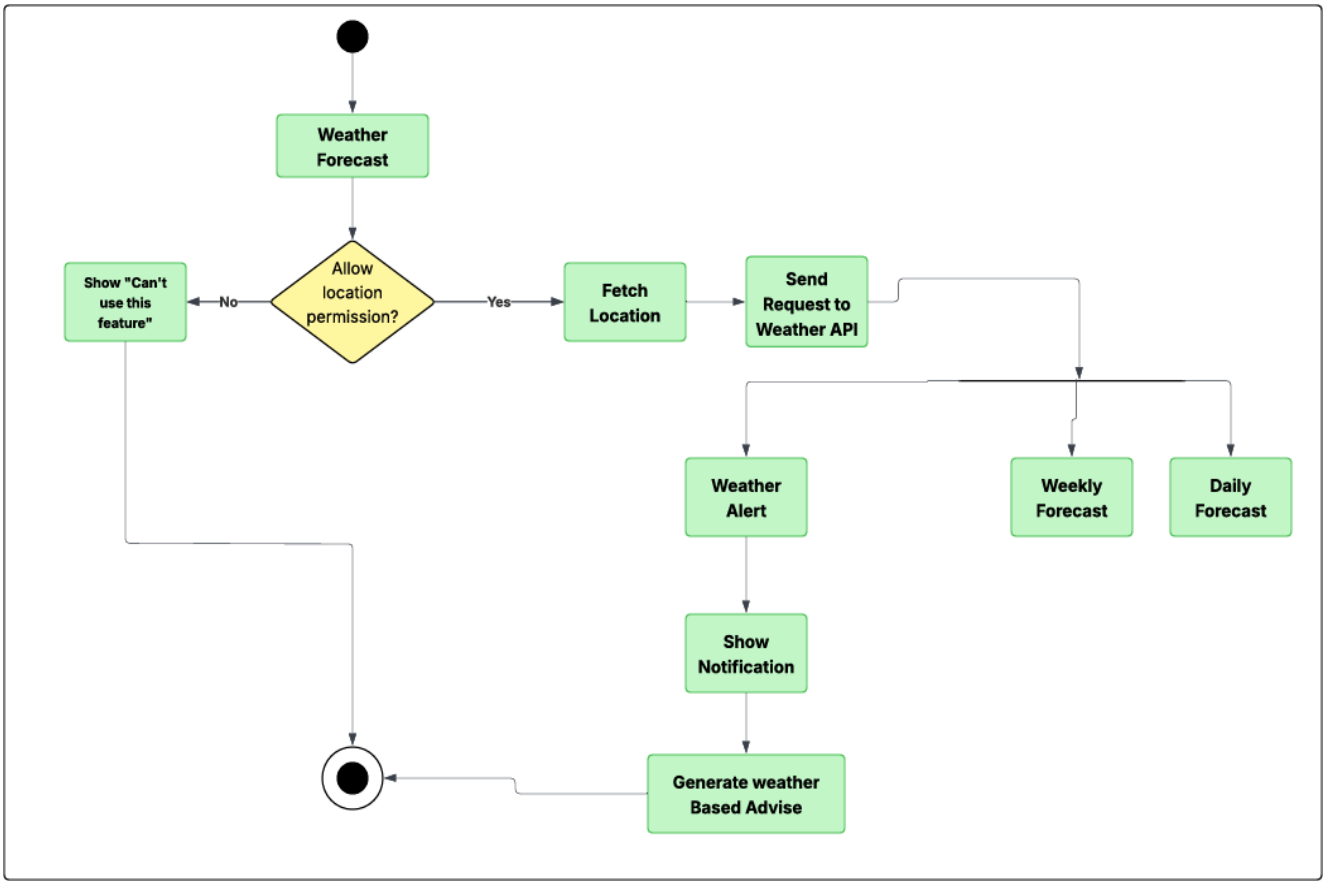
User Account and Language service



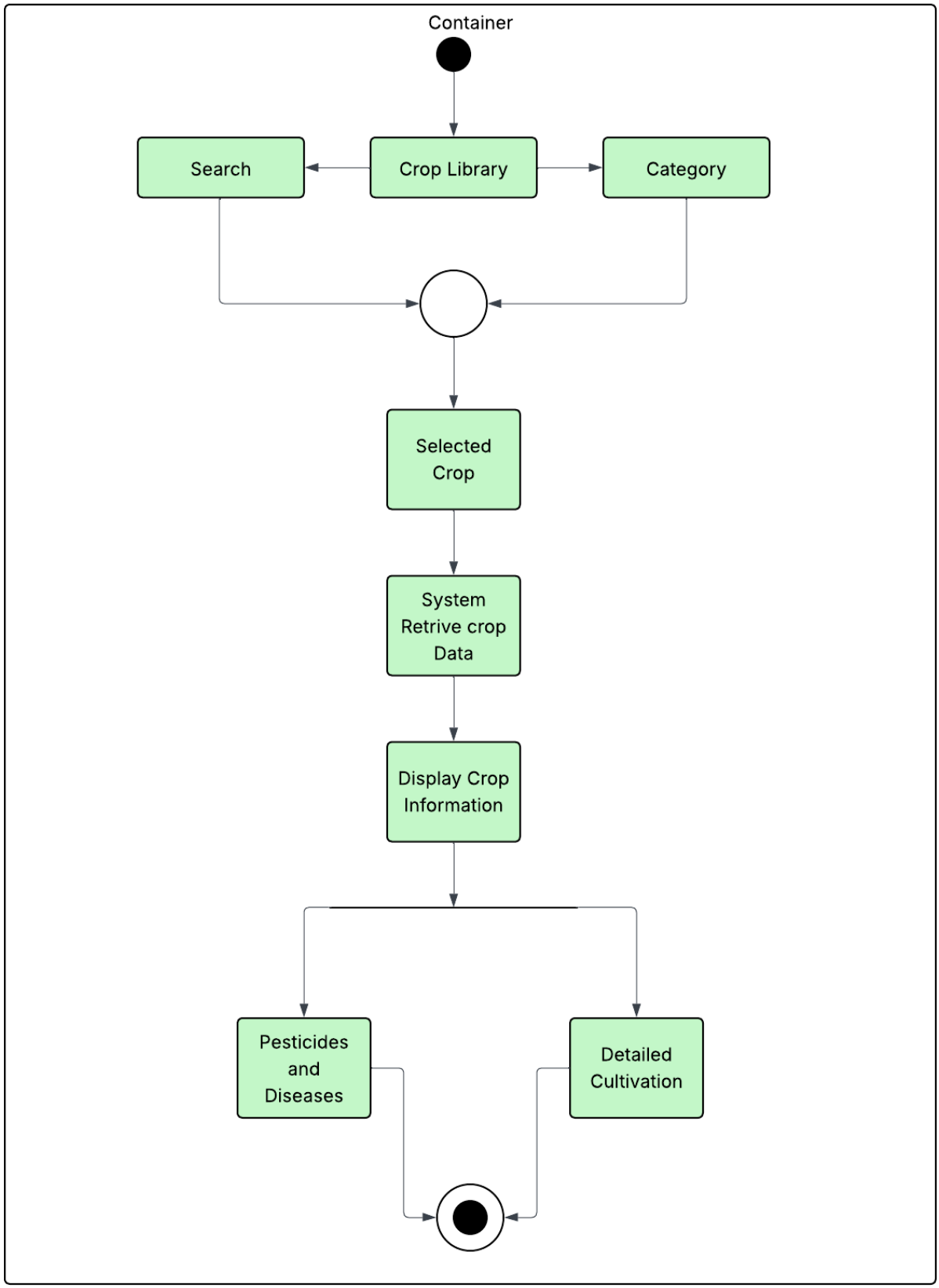
Crop Health Analysis



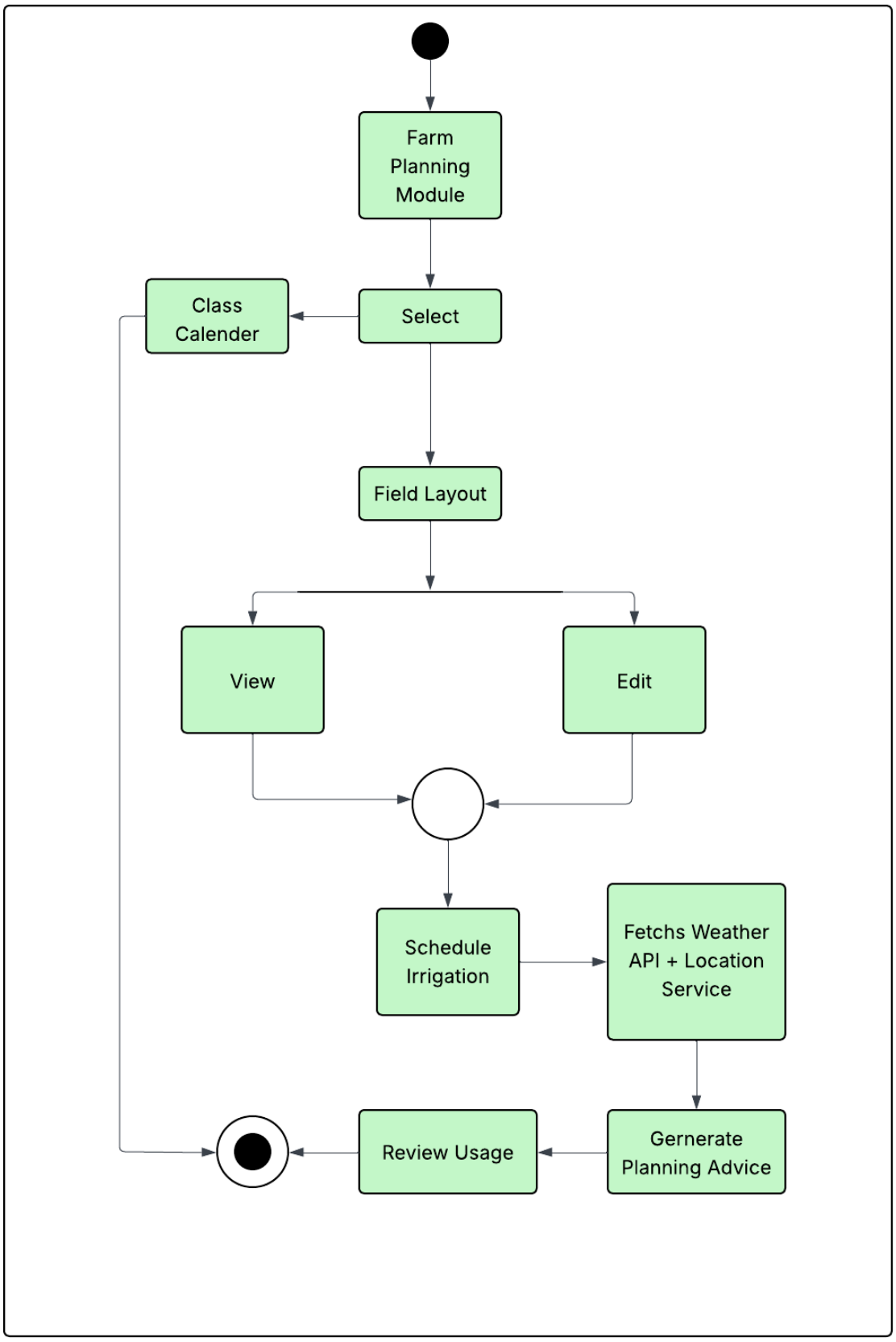
Weather Forecast



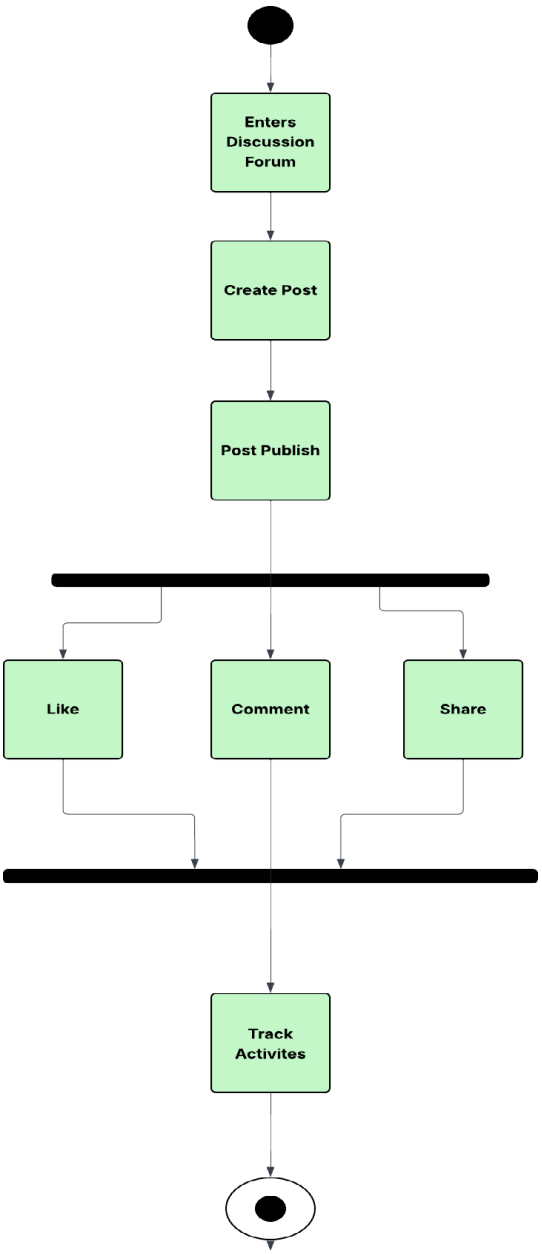
Crop Library



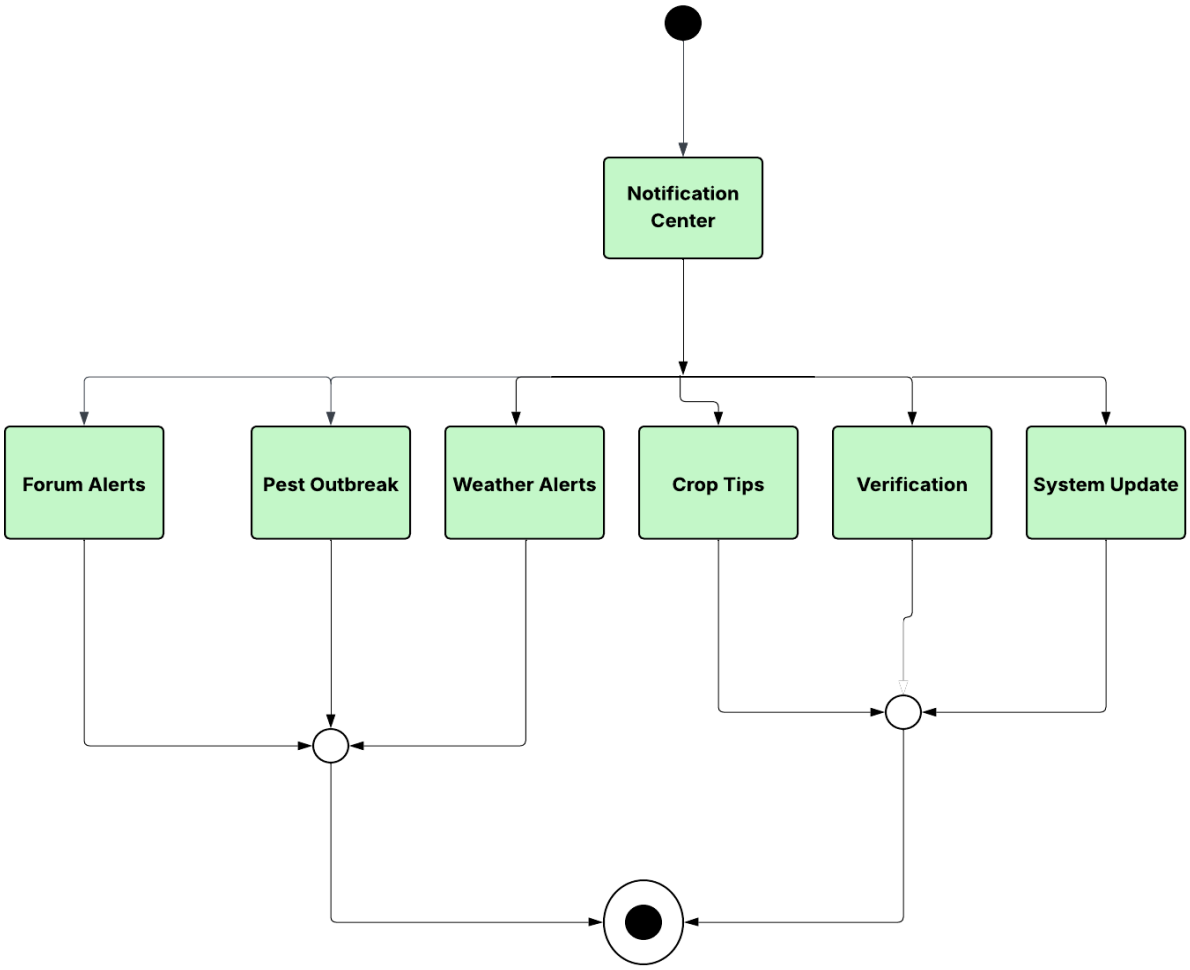
Farm Planning



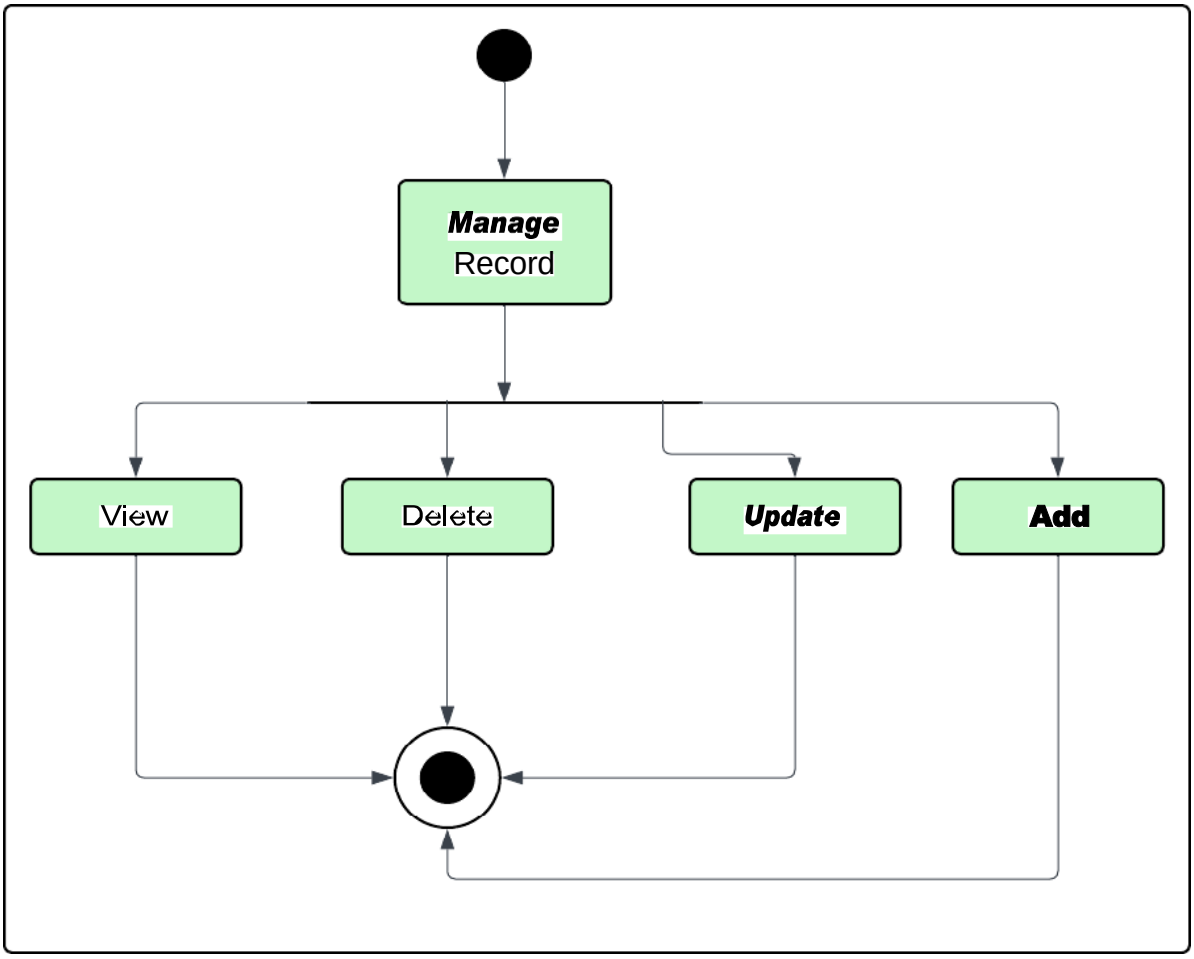
Discussion Forum

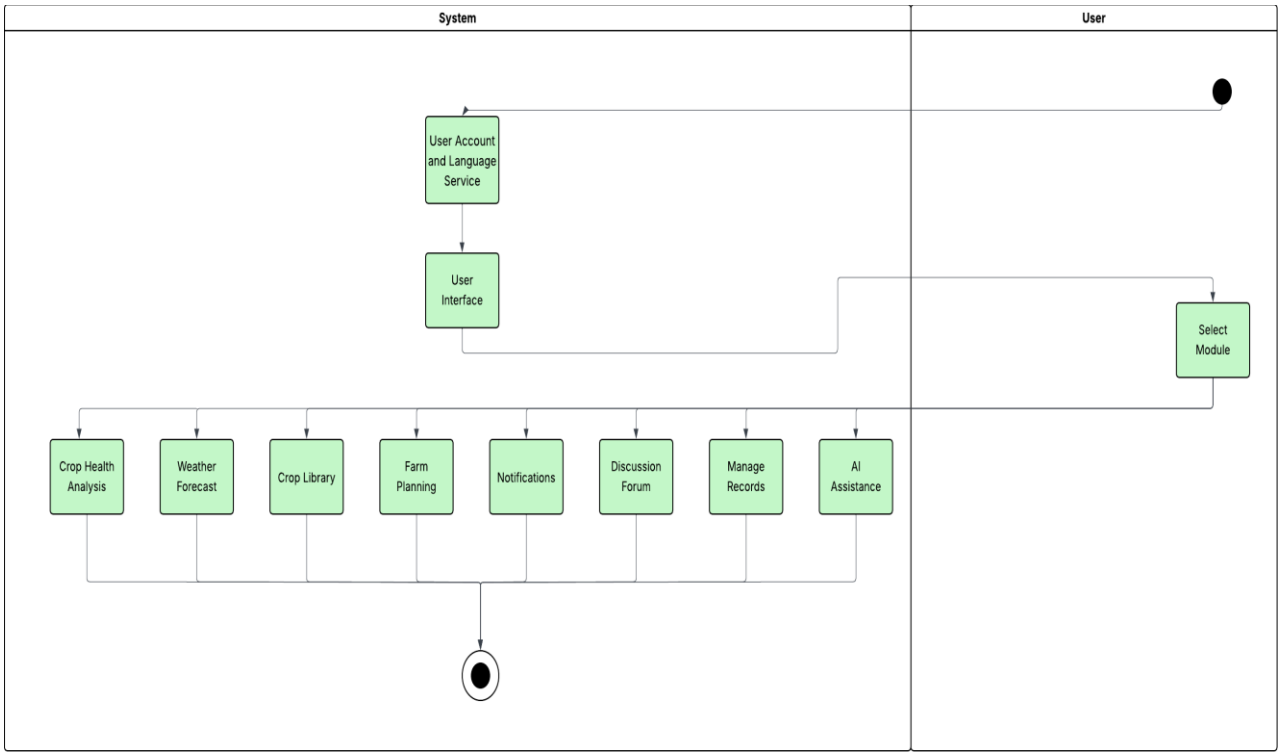


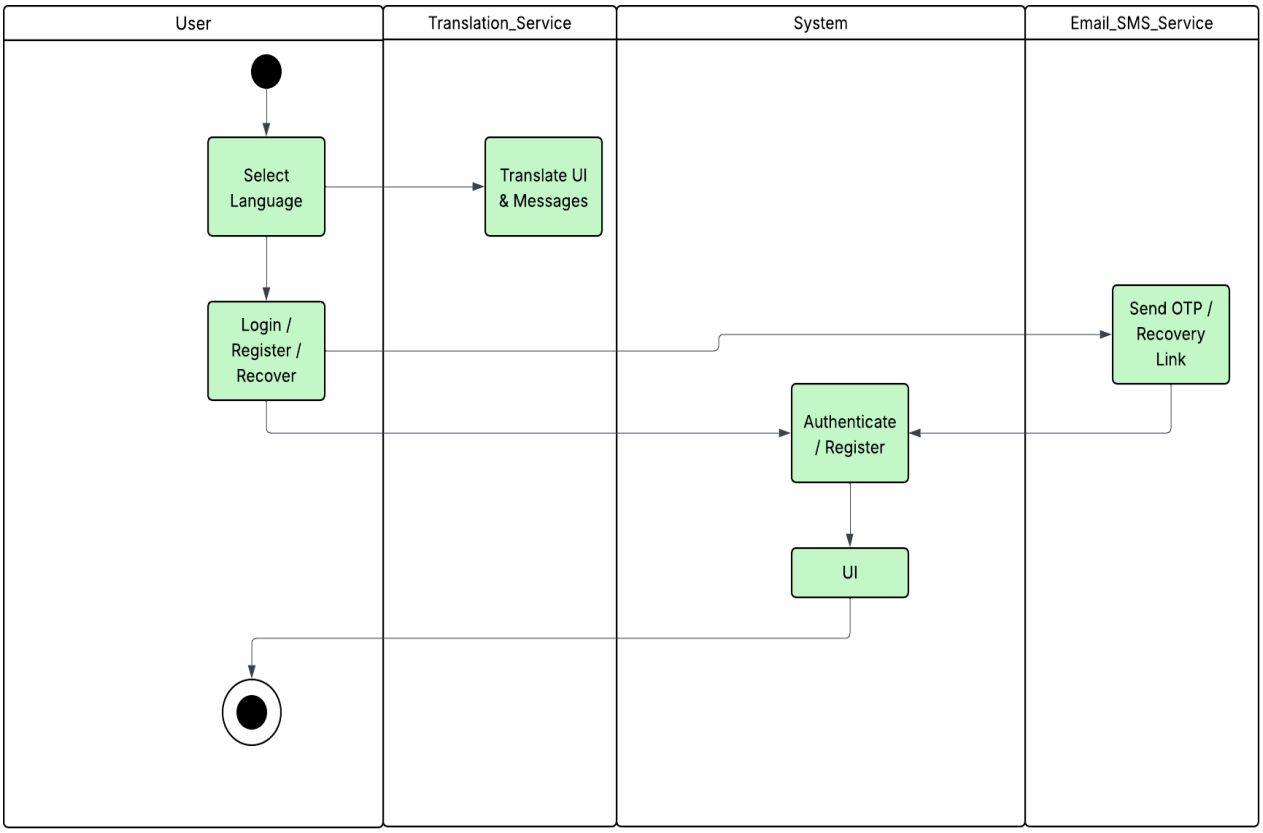
Notifications

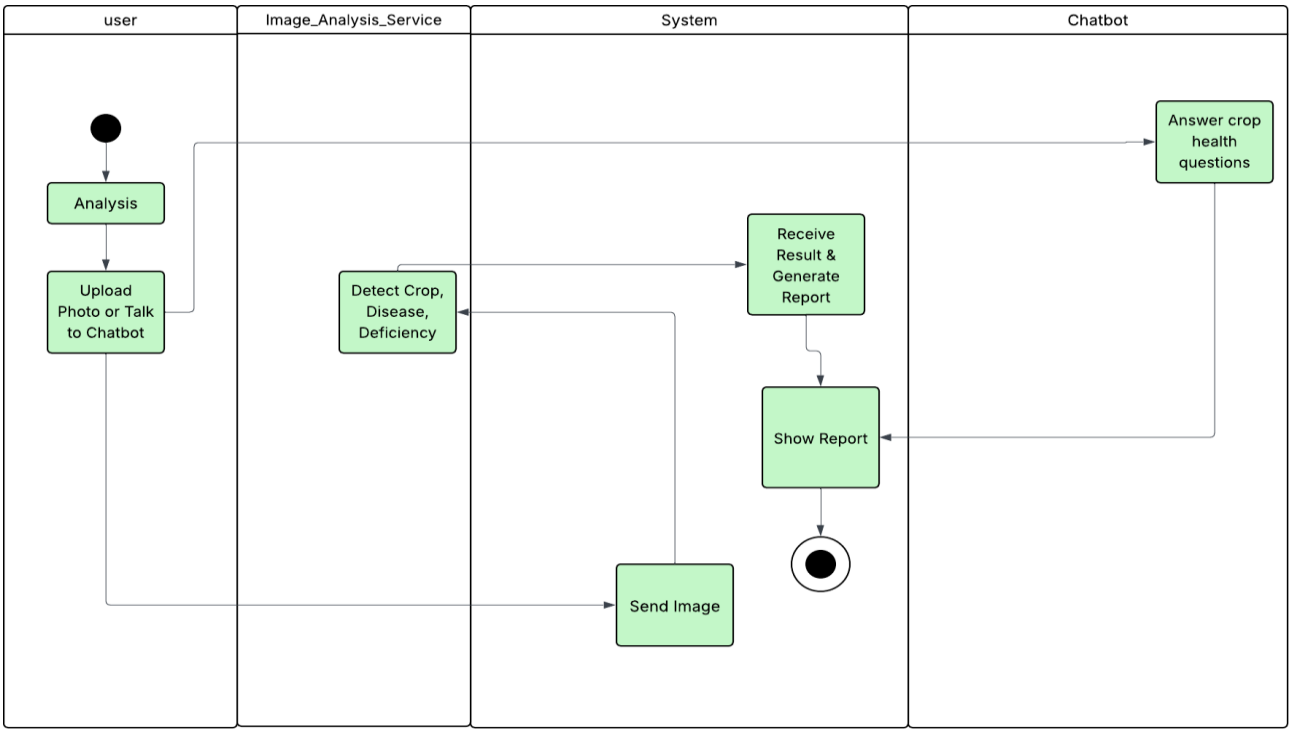


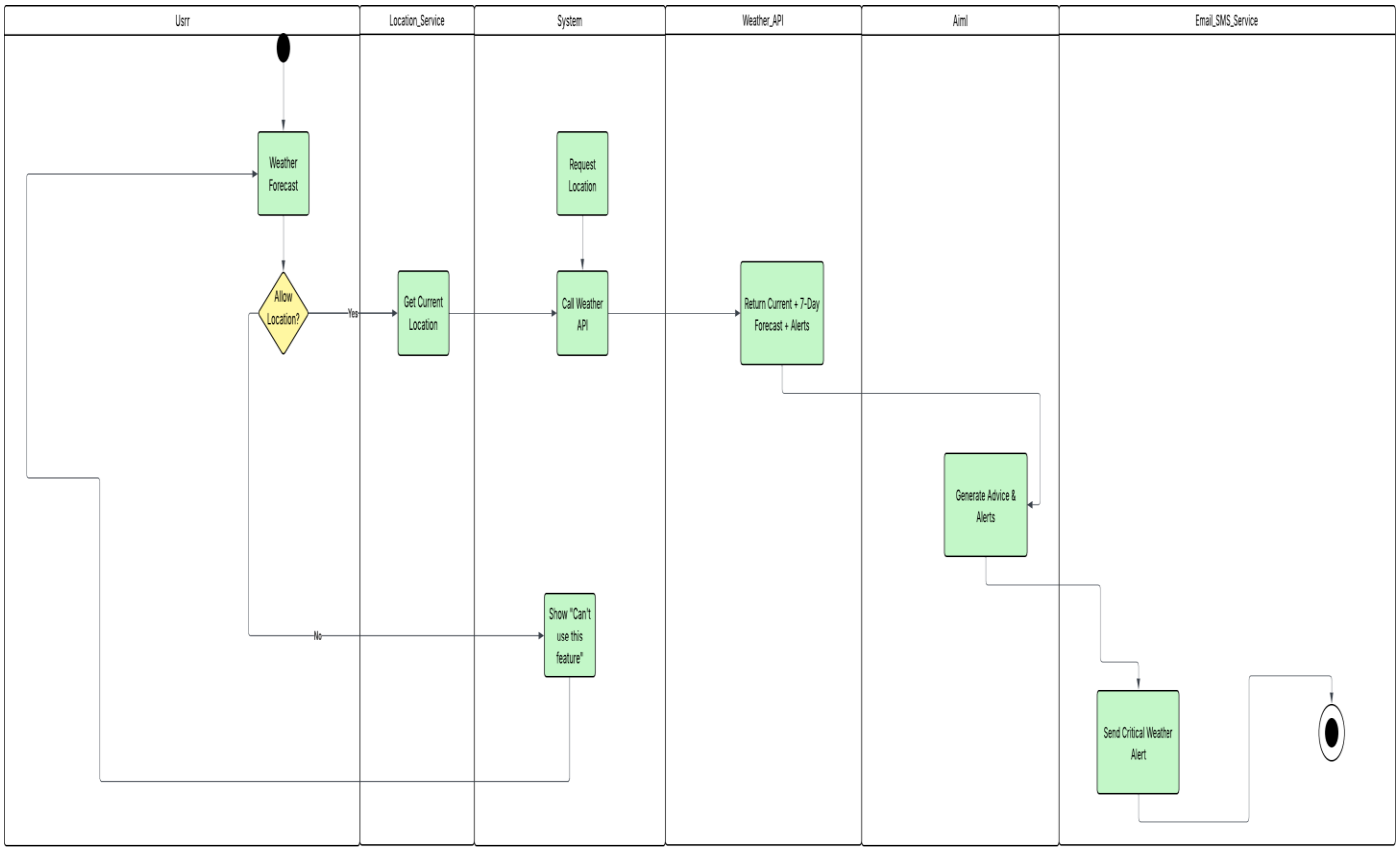
Manage Records

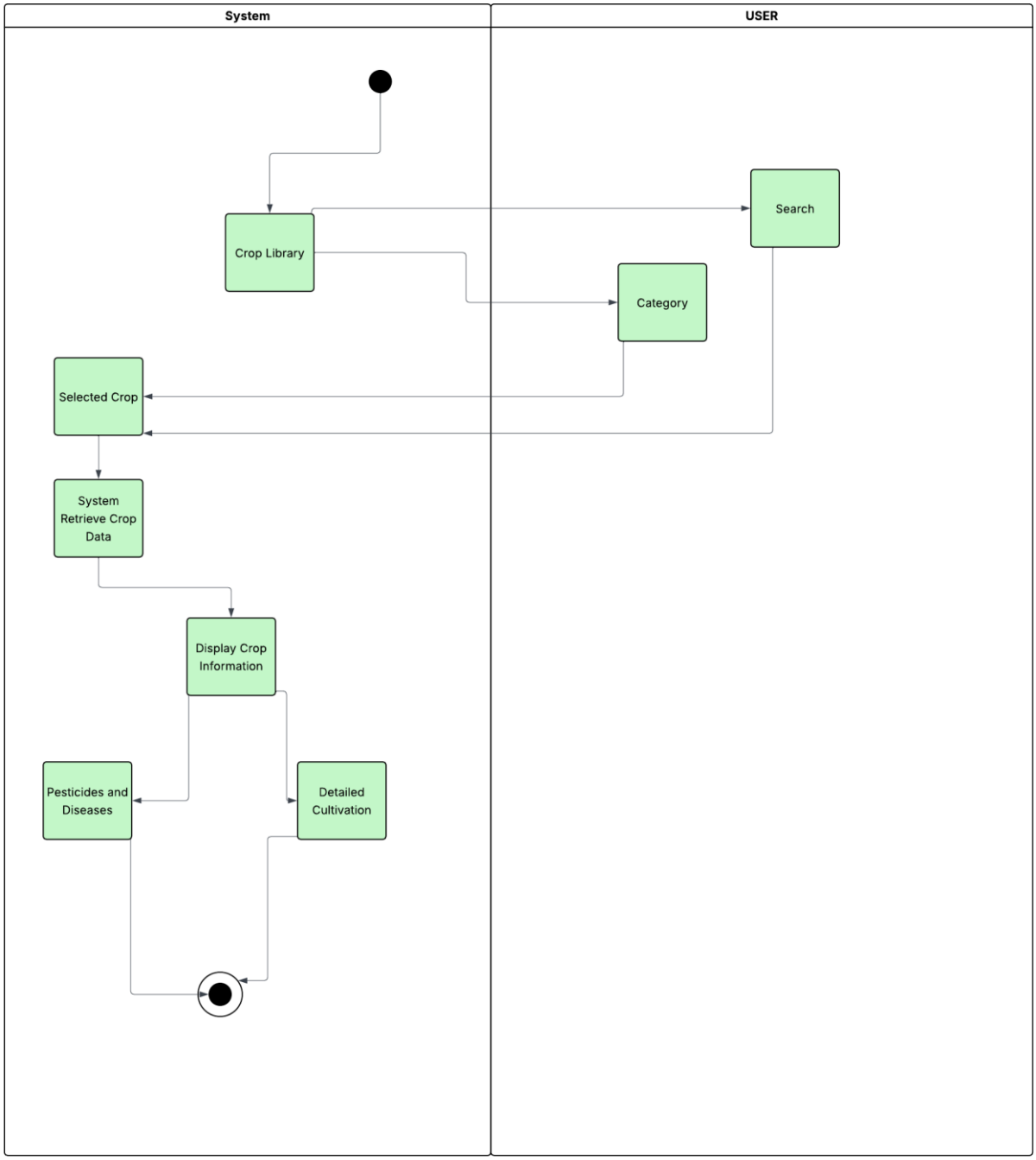


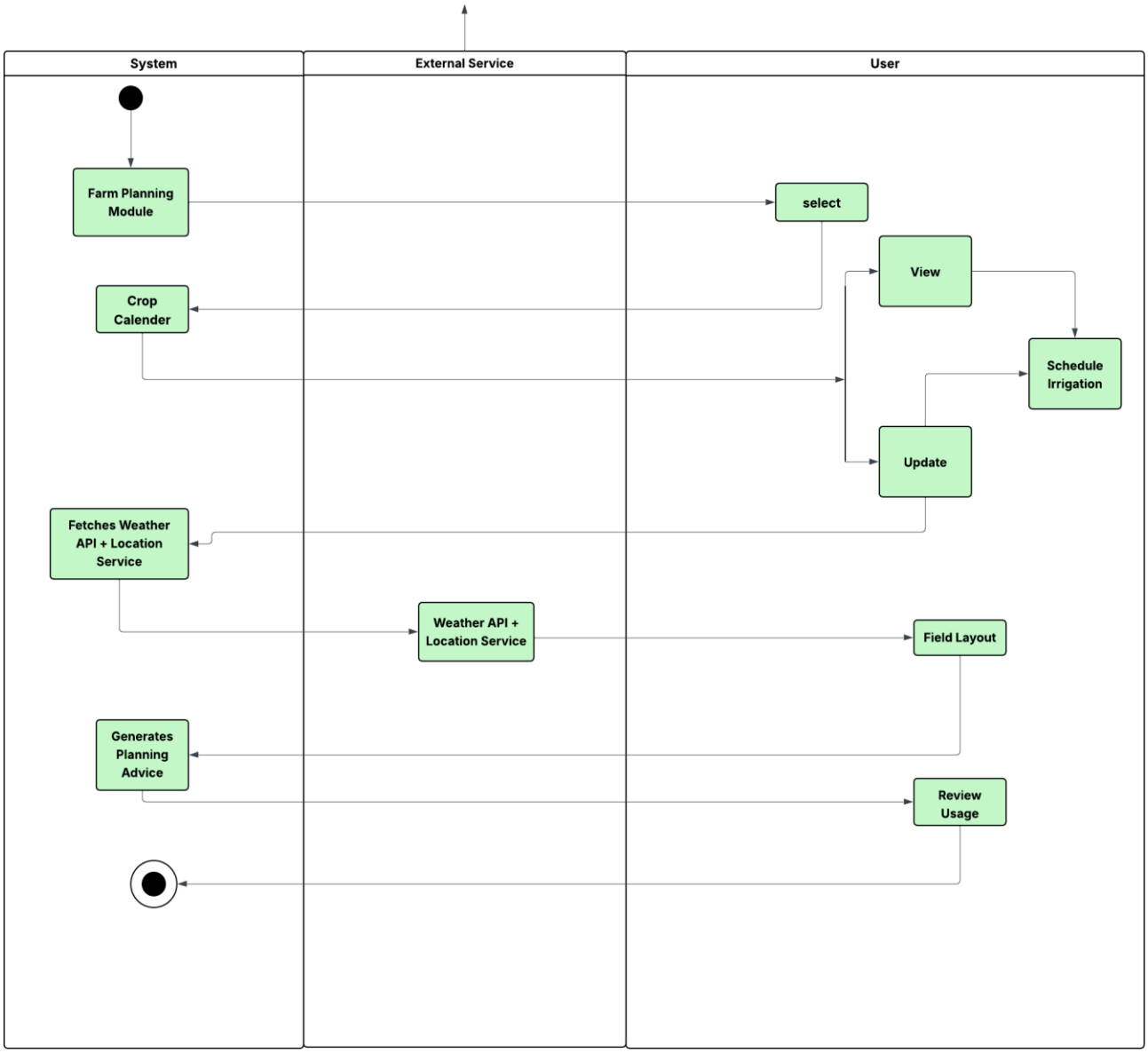

System
User
User Account and Language Service
User Interface
Select Module
Crop Health Analysis
Weather Forecast
Crop Library
Farm Planning
Notifications
Discussion Forum
Manage Records
AI Assistance

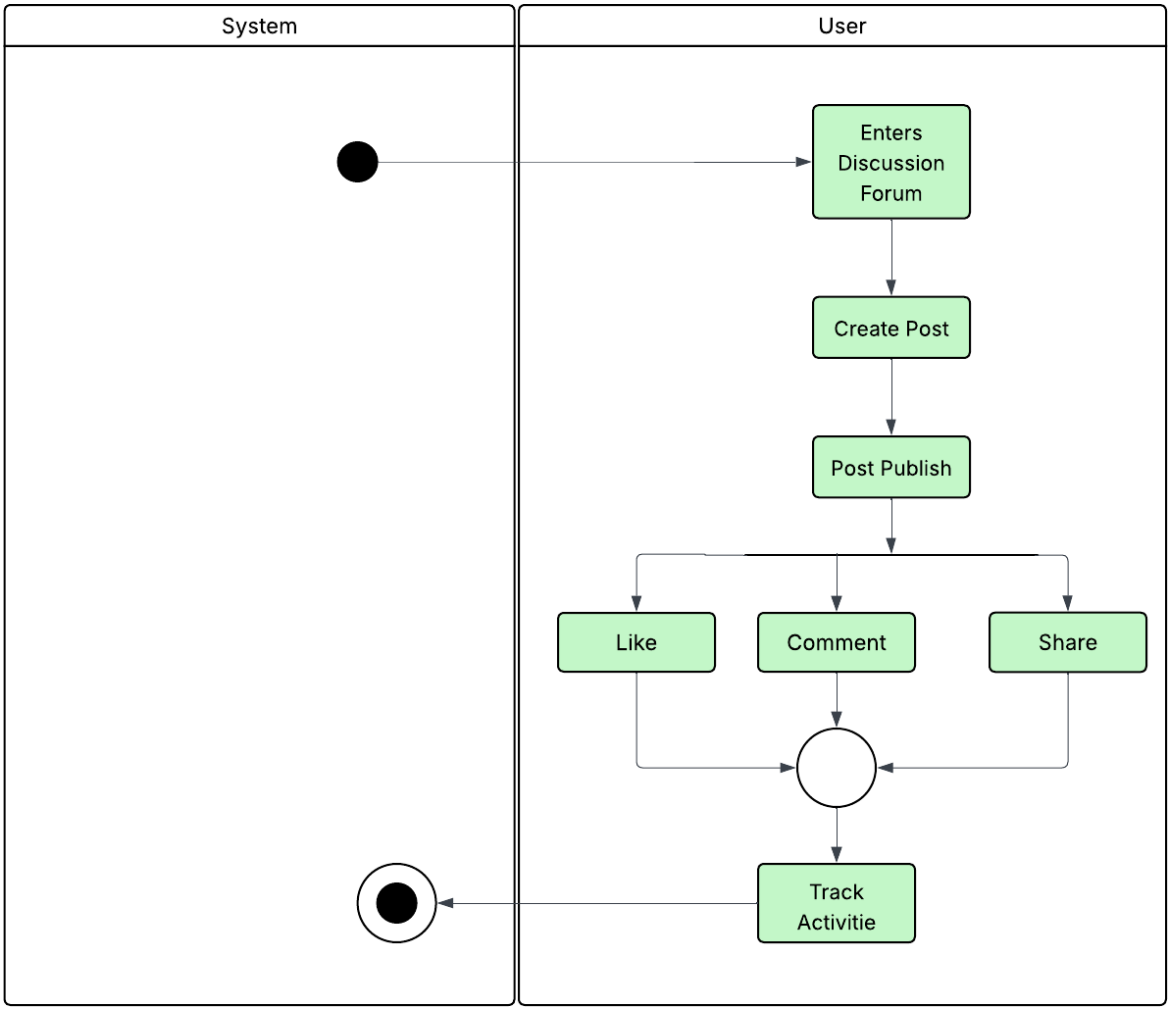

User
Translation_Service
System
Email_SMS_Service
Select Language
Translate UI & Messages
Login / Register / Recover
Send OTP / Recovery Link
Authenticate / Register
UI

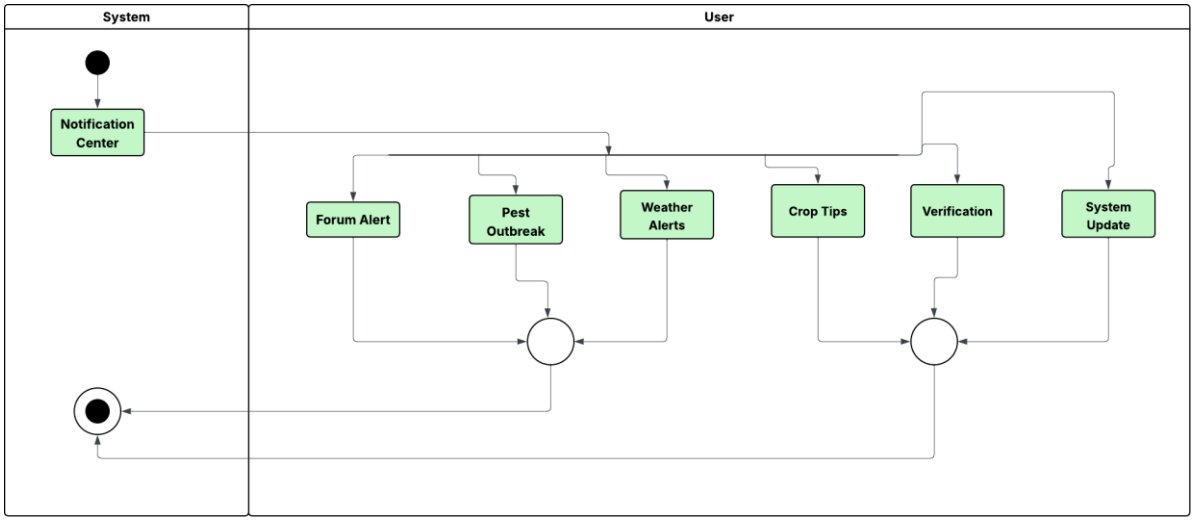

user
Image_Analysis_Service
System
Chatbot
Analysis
Upload Photo or Talk to Chatbot
Detect Crop, Disease, Deficiency
Receive Result & Generate Report
Answer crop health questions
Show Report
Send Image

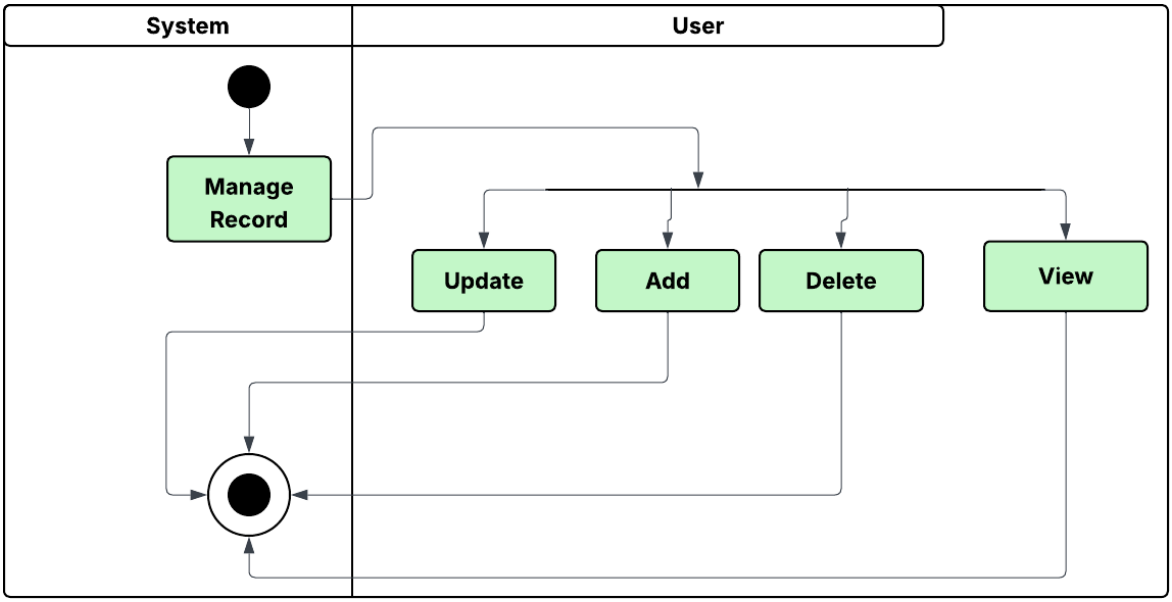

Usrr
Location_Service
System
Weather_API
Aiml
Email_SMS_Service
Weather Forecast
Allow Location?
Yes
No
Get Current Location
Request Location
Call Weather API
Return Current + 7-Day Forecast + Alerts
Generate Advice & Alerts
Show "Can't use this feature"
Send Critical Weather Alert


System
USER
Crop Library
Search
Category
Selected Crop
System Retrieve Crop Data
Display Crop Information
Pesticides and Diseases
Detailed Cultivation

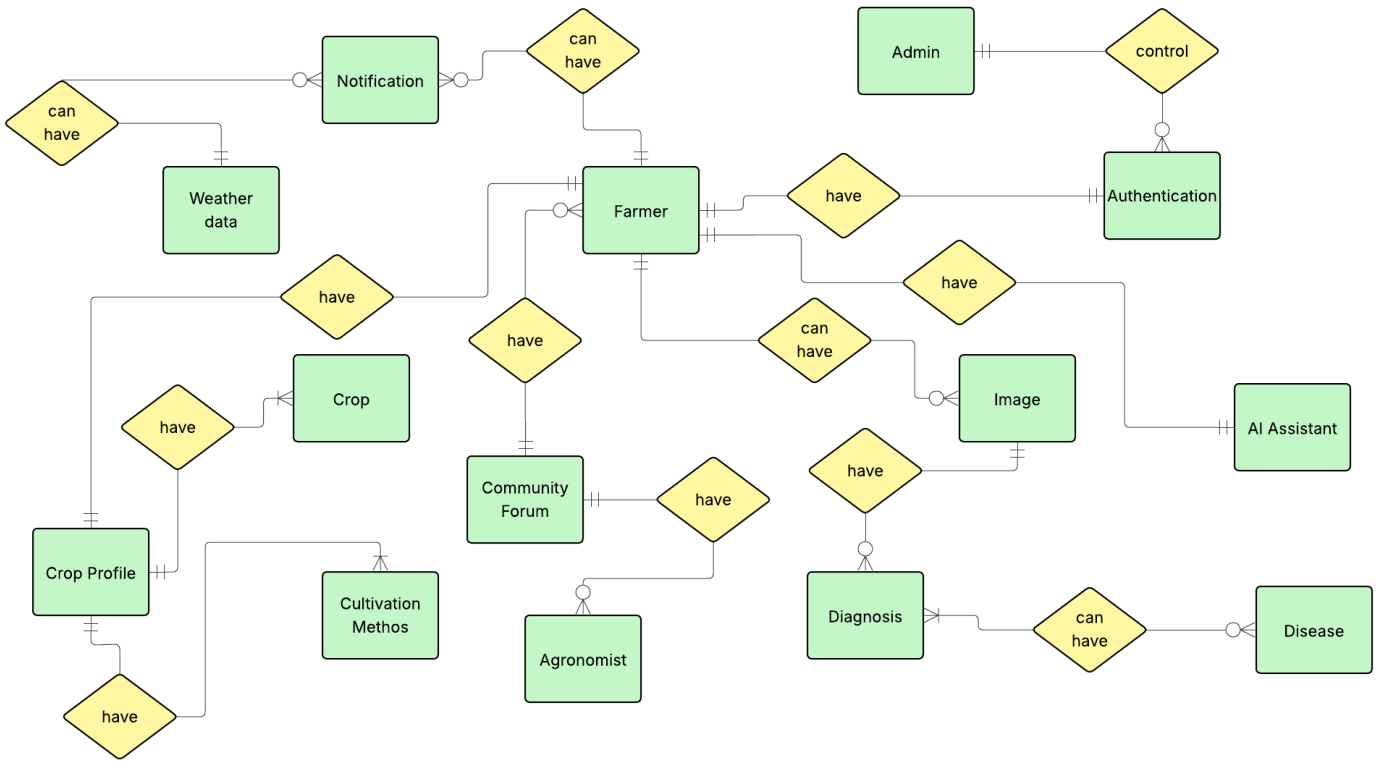

System
External Service
User
Farm Planning Module
select
View
Crop Calender
Schedule Irrigation
Update
Fetches Weather API + Location Service
Weather API + Location Service
Field Layout
Generates Planning Advice
Review Usage


System
User
Enters Discussion Forum
Create Post
Post Publish
Like
Comment
Share
Track Activitie


System
User
Notification Center
Forum Alert
Pest Outbreak
Weather Alerts
Crop Tips
Verification
System Update


System
User
Manage Record
Update
Add
Delete
View

ER Diagram


Notification
can have
can have
Weather data
Admin
control
Authentication
Farmer
have
have
have
have
can have
Image
AI Assistant
Crop
have
Community Forum
have
have
Crop Profile
Cultivation Methos
Agronomist
Diagnosis
can have
Disease
have